

Exploring experimental research: Methodologies, designs, and applications across disciplines

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Abstract

Experimental research is a primary scientific method used to elucidate cause-and-effect correlations between variables in diverse fields. Specifically, this paper outlines the essential characteristics of experimental research, encompassing variable manipulation, controlled settings, random assignment, and precise measurement procedures to enable causal inferences. Moreover, the paper clarifies many experimental designs, including randomized controlled trials, true experimental designs, quasi-experimental designs, and single-case designs, each adapted to certain research contexts. Subsequently, the paper delineates the procedural processes for doing experimental research, underscoring the significance of methodological rigor from study design to result interpretation. Furthermore, it outlines the possible challenges to internal and external validity, emphasizing the importance of addressing confounding factors to ensure strong experimental results. In addition, the paper examines the timing of pre-tests and post-tests, the complexities of experimental design, and emerging trends such as internet-based experiments and ex post facto research. Finally, this paper seeks to furnish scholars with a detailed comprehension of experimental inquiry through an extensive analysis of research procedures, designs, and applications across several academic fields.

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1. Introduction

Experimental research is a scientific method used to investigate cause-and-effect relationships between variables (Frissen & Evans, 2024; Sturmberg & Marcum, 2024). In experimental research, the researchers manipulate one or more independent variables to observe the effect on a dependent variable, while controlling for extraneous variables that could influence the results (Cohen et al., 2017; Mattila et al., 2021).

Experimental research is characterized by key features such as the manipulation of independent variables, control of extraneous variables, random assignment of participants, and the establishment of cause-and-effect relationships (Frissen & Evans, 2024; Rauteda, 2025). Below are more detailed explanations.

Manipulation of variables: The term “manipulation of variables” refers to the process by which a researcher purposefully alters or manipulates various independent variables in order to examine the effect that these changes have on the dependent variable (Thomas, 2021). Because of this alteration, researchers are able to determine the causal relationship (Van der Stede, 2014).

Controlled conditions: In order to guarantee that the outcomes of the experiment are not influenced by any extraneous variables, experimental research is carried out under controlled settings (Street, 1995). It is possible for researchers to isolate the effects of the independent variable on the dependent variable if they adjust for these variables (Ross & Morrison, 2013).

Random assignment: The participants in experimental research are often randomized to various groups by a random assignment process (Shadish et al., 2008). This is done to ensure that each group is comparable at the beginning of the study before the experiment begins (Edgington, 1985). According to Rogers and Revesz (2019), random assignment is an effective method for reducing the impact of individual variations and prior biases.

Measurement of variables: Evaluation of the relationship between the variables in experimental research, it is necessary to measure both the independent and dependent variables in order to evaluate the relationship between them (Sigall & Mills, 1998). Accurate data collection is ensured by the use of measurement tools that are both valid and trustworthy (Bhandari, 2022).

Causal inferences: One of the fundamental objectives of experimental research is to determine the causal linkages that exist between the variables being studied (Rutter, 2007). In order for researchers to draw conclusions regarding the nature of the link between the variables, they must first manipulate the independent variable and then observe the effect that this manipulation has on the dependent variable, all the while controlling for any extraneous variables (Kuang et al, 2020).

Accordingly, Leppink (2019) noted that research that is conducted through experimentation is utilized extensively in a variety of sectors, including psychology, education, medicine, and the natural sciences. Researchers are able to put hypotheses to the test, make predictions, and contribute to our understanding of how variables interact with one another and influence behavior, outcomes, or phenomena.

2. Randomized controlled trials (RCTs)

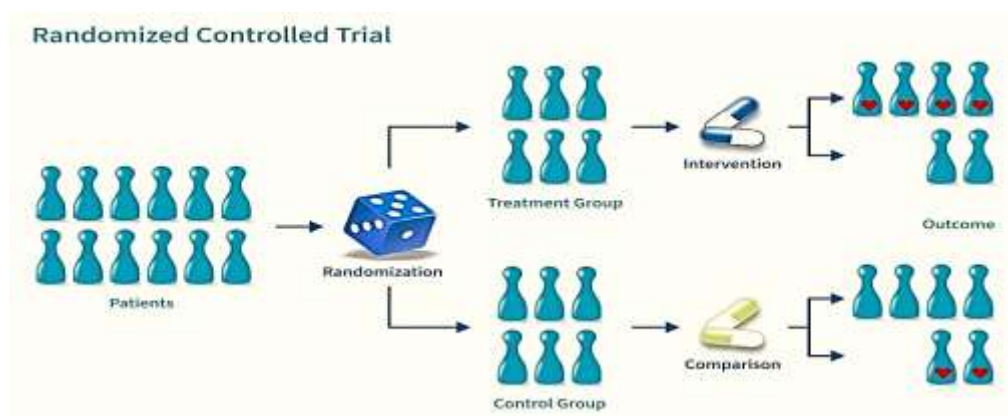


Figure 1. Randomized controlled trials (RCTs). Source: Simkus (2024).

Explanation: Experimental designs known as randomized controlled trials (RCTs) involve randomly assigning people to various groups. In these investigations, at least one group receives the intervention or treatment that is being examined, while another group acts as a control. According to Ross and Morrison (2013), random assignment helps to ensure that the groups are comparable, which in turn enables researchers to draw conclusions about the effects of the intervention based on causality that they have established.

Example: Students are randomly assigned to either an experimental group, which is given Method A, or a control group, which is given Method B, in a study that compares the efficacy of two different teaching approaches for enhancing students' mathematical skills. After that, the results of the math tests are compared between the two groups in order to evaluate the effectiveness of the instructional strategies.

3. Designs in educational experiments

Explanation: Various experimental designs are utilized in educational research to investigate the efficacy of various teaching approaches or interventions. This word refers to the various experimental designs that are utilized. The level of control that these designs have on the variables and their capacity to draw conclusions about causal relationships are both different (e.g., Chan et al., 2021).

Example: Randomized controlled trials, quasi-experimental designs, and single-case designs are examples of the types of experimental designs that are frequently utilized in educational research programs.

4. True experimental designs

Explanation: The true experimental designs entail randomly allocating people to various groups and altering the independent variable in order to evaluate the influence that it has on the variable that is being studied (the dependent variable). According to Ross and Morrison (2013), these designs make it possible for researchers to establish links between causes and effects.

Example: Students are randomly assigned to either an experimental group, which is given the new way of instruction, or a control group, which is given the traditional style of instruction, in a study that investigates the impact of a new teaching approach on reading comprehension. Following this, the scores of both groups are compared with regard to their reading comprehension.

5. Quasi-experimental designs

Explanation: The difference between quasi-experimental designs and true experimental designs is that the former do not include random assignment to groups. According to Ross and Morrison (2013), individuals are not randomly assigned to groups but rather are randomized to groups based on certain qualities or conditions.

Example: In a study that investigates the efficacy of a tutoring program for enhancing academic performance, it is possible to compare the academic performance of students who willingly participate in the program (the treatment group) with the performance of students who do not participate in the program (the control group). Random assignment is not possible because participants choose their own groups at the beginning of the study.

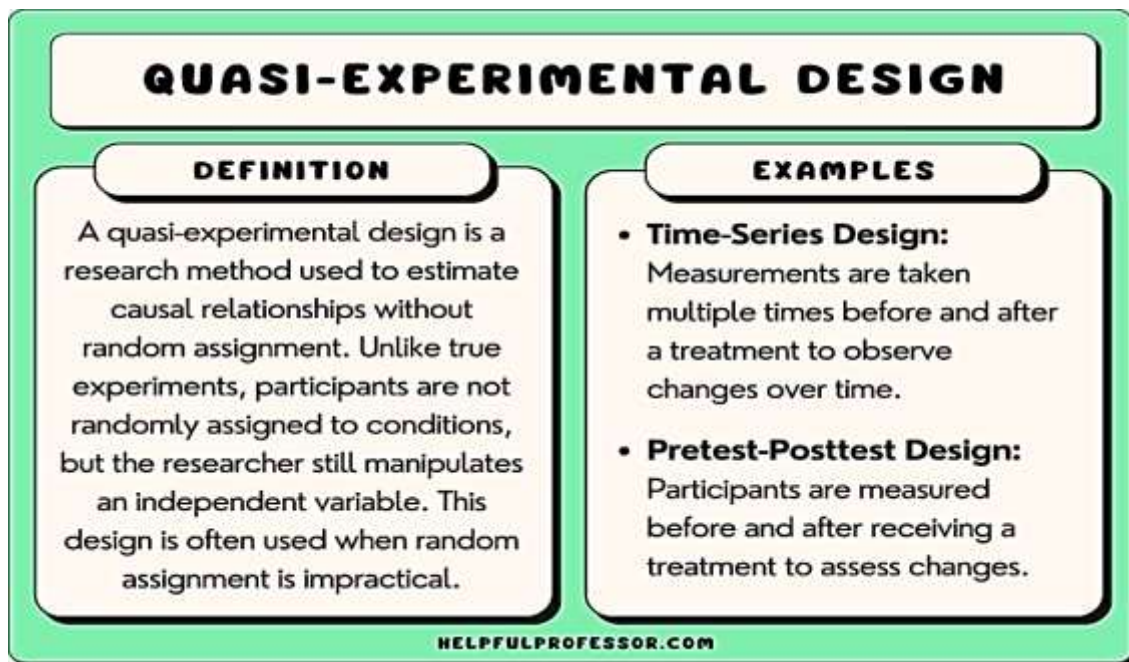


Figure 2. Quasi-experimental designs. Source: Cornell (2024).

6. Single-case ABAB design

Explanation: In the field of applied behavior analysis, one sort of experimental design that is utilized is known as the single-case ABAB design. It entails alternating between baseline phases (A), in which the intervention is not applied, and intervention phases (B), in which the intervention is implemented, in order to evaluate the effects of the intervention (Kazdin, 2021).

Example: An investigation of the efficacy of a behavior modification program for reducing disruptive behaviors in a student might involve alternating between periods of baseline observation (no intervention) and intervention implementation. This would indicate that the program is effective in reducing disruptive behaviors. It is possible for researchers to determine whether or not the intervention results in changes in behavior thanks to the design.

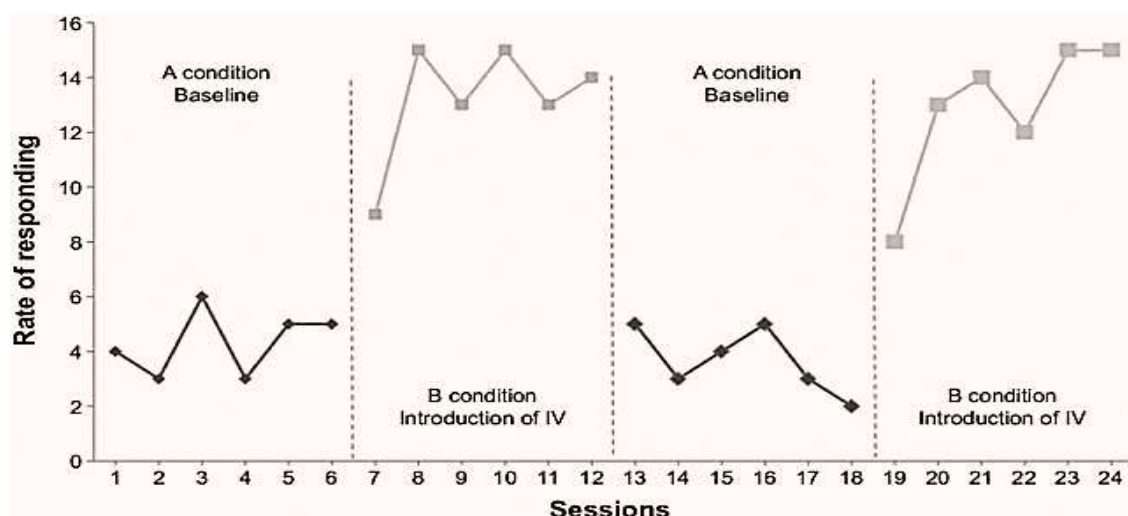


Figure 3. Single-case ABAB design. Source: Steingrimsdottir and Arntzen (2015).

7. Procedures in conducting experimental research

Explanation: According to Ross and Morrison (2013), the following are the systematic stages and methodologies that researchers take when conducting experimental research. These include designing the study, choosing participants, executing treatments, collecting data, and interpreting the outcomes.

Example: The researchers may begin by designing the study, recruiting participants, randomly assigning them to groups, implementing the teaching method, administering pre- and post-tests, collecting and analyzing data, and drawing conclusions based on the findings. This is all part of the process of conducting an experiment to evaluate the effectiveness of a new teaching method.

8. Threats to internal and external validity in experiments

Explanation: The term “internal validity” refers to the extent to which the observed effects may be attributed to the independent variable rather than confounding variables. On the other hand, the term “external validity” refers to the extent to which the findings can be generalized to other populations or contexts. According to Andrade (2018), various threats to validity have the potential to impair the validity of experimental data.

Example: When participants in various groups are systematically different at the beginning of the investigation, this is an example of selection bias, which is a common danger to the internal validity of the study. The utilization of a highly controlled laboratory environment that does not accurately reflect the conditions that exist in the actual world could pose a risk to the external validity of the study.

9. The timing of the pre-test and the post-test

Explanation: It is common practice in experimental research to administer pre-tests prior to the intervention in order to evaluate the baseline levels of the dependent variable. Post-tests, on the other hand, are administered after the intervention in order to evaluate the changes that have occurred in the dependent variable. For the purpose of determining whether or not the intervention was successful, the timing of these tests is very significant (Marsden & Torgerson, 2012).

Example: It is possible that students will first be required to take a math test, often known as a pre-test, in order to establish their baseline performance in a study that is testing the impact of a new teaching approach on mathematical skills. A second arithmetic test, known as a post-test, would be administered to the individuals after they had received the intervention in order to evaluate any gains in their abilities.

10. The design experiment

Explanation: It is possible that this term is referring to the process of creating an experiment, which includes establishing the research issue, selecting the proper design, and organizing the implementation and analysis of the study (Mattila et al., 2021).

Example: The first step for researchers who are interested in investigating the effects of a mindfulness-based intervention on the well-being of students is to design the experiment, define the research question, choose the experimental design (for example, a randomized

controlled trial), and plan the specifics of the intervention as well as the procedures for collecting data.

11. Internet-based experiments

Explanation: The experiments in question are carried out online, and they frequently make use of web-based platforms in order to recruit participants, carry out interventions, and collect data. According to Gavras et al. (2007), internet-based experiments provide a number of benefits, including the ability to access a wide range of people and the cost-effectiveness of the research. However, these experiments may also encounter issues in terms of sample representativeness and data security.

Example: An investigation into the effects of an online mindfulness training program on stress levels could involve the recruitment of participants through social media or online marketing, the delivery of the intervention through a web-based platform, and the collection of data through online surveys or evaluations.

12. Ex post facto research

Explanation: In the context of research that is carried out after the fact, this refers to situations in which the researcher does not have control over the independent variable. The study of naturally occurring variations between groups is a common component of ex post facto research, as opposed to the manipulation of factors (Rohwer, 2022).

Example: In the event that a researcher is interested in investigating the ways in which gender influences academic achievement, they might examine the data that is already available from school records and compare the academic performance of male and female students. Because the researcher is unable to influence the gender of the participants, this kind of research is classified as ex post facto.

13. Conclusion

In conclusion, experimental research is an essential component of scientific investigation because it provides researchers with a potent instrument for revealing causal linkages, putting hypotheses to the test, and expanding knowledge across a variety of fields. When researchers embrace methodological rigor, make use of a variety of experimental designs, and address validity risks, they are able to harness the full potential of experimental research and produce contributions that are important to their respective areas. Researchers have a responsibility to maintain vigilance in their pursuit of rigorous and meaningful scientific inquiry as experimental procedures continue to develop and adapt to shifting research contexts.

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Declaration of AI usage

The author used Grammarly to polish the language in some parts of the article. However, he declared that the content of the article is totally his.

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